

Section I 10 marks**Attempt Questions 1-10****Allow about 15 minutes for this section**

1. Which of the following is the angle between the vectors $\vec{a} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$
 - A. 10°
 - B. 63°
 - C. 140°
 - D. 117°

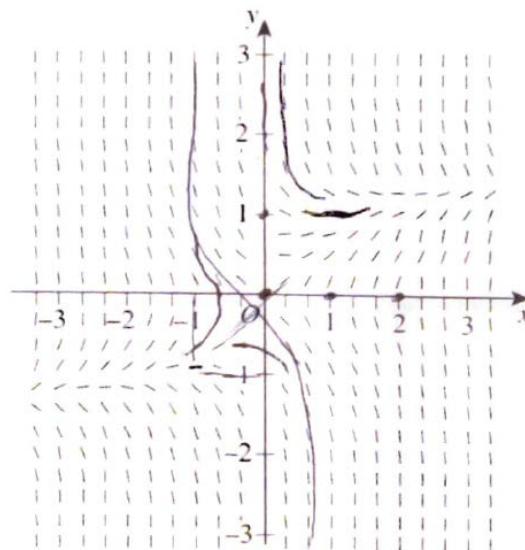
2. Which of the following equal to $\cos \theta$
 - A. $\sin^2 \frac{\theta}{2} - \cos^2 \frac{\theta}{2}$
 - B. $2 \cos^2 \frac{\theta}{2} + 1$
 - C. $1 - 2 \cos^2 \theta$
 - D. $\frac{\sin \theta}{\tan \theta}$

3. The polynomial $P(x) = x^3 - x^2 - 5x - 3$ has a double root at $x = \alpha$. What is the value of α ?
 - A. $-\frac{5}{3}$
 - B. -1
 - C. 1
 - D. $\frac{5}{3}$

4. Which of the following is equivalent to $\int \sin x \cos x \, dx$?
 - A. $-\cos 2x + c$
 - B. $-\frac{1}{2} \cos 2x + c$
 - C. $-\frac{1}{4} \cos 2x + c$
 - D. $\frac{1}{4} \cos 2x + c$

5. What is the domain of $y = 3 \cos^{-1}(2x + 1)$?
 - A. Domain $[-1, 0]$
 - B. Domain $(1, 0)$
 - C. Domain $[-\frac{1}{2}, 0]$
 - D. Domain $[\frac{1}{2}, 0]$

6. In a Mathematics class a teacher can award one of 4 grades, A, B, C or D to each student. What is the minimum number of students required so that at least 8 students are guaranteed to receive the same grade?
- A. 28
B. 29
C. 32
D. 33
7. Which of the following is the range of the function $\frac{1}{x^2 + 1}$
- A. $(-\infty, \infty)$
B. $(-\infty, 1]$
C. $(0, 1]$
D. $[0, 1]$
8. A curve has an asymptote at $x = \frac{\pi}{3}$. Which of the following could be the equation of the curve?
- A. $y = \sec\left(x - \frac{\pi}{3}\right)$
B. $y = \sec\left(x + \frac{\pi}{3}\right)$
C. $y = \cot\left(x - \frac{\pi}{3}\right)$
D. $y = \operatorname{cosec}\left(x + \frac{\pi}{3}\right)$
9. Which of the following is the primitive of $\frac{4}{\sqrt{9 - x^2}}$?
- A. $\frac{4}{3} \sin^{-1} \frac{x}{3} + c$
B. $\frac{4}{3} \sin^{-1} 3x + c$
C. $4 \sin^{-1} 3x + c$
D. $4 \sin^{-1} \frac{x}{3} + c$
10. Which of the following best represents the differential equation shown in the slope field?
- A. $\frac{dy}{dx} = \frac{x}{y} - y^2$
B. $\frac{dy}{dx} = \frac{x}{y} + y^2$
C. $\frac{dy}{dx} = -\frac{x}{y} - y^2$
D. $\frac{dy}{dx} = -\frac{x}{y} + y^2$



Section II 60 marks**Attempt Questions 11-14****Allow about 1 hour and 45 minutes for this section**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available
For questions in Section II, your response should include relevant mathematical reasoning
and/or calculations

Question 11 (15 marks) Use a SEPARATE writing booklet

- (a) Solve $\frac{3}{2x-5} \leq -1$ **3**
- (b) Find the derivative of $\cos^{-1}\left(\frac{3x}{2}\right)$ **1**
- (c) For what values of m are the distinct vectors $\begin{pmatrix} m \\ 2m+6 \end{pmatrix}$ and $\begin{pmatrix} m+1 \\ -1 \end{pmatrix}$ perpendicular? **2**
- (d) Find $\int \cos^2 4x \, dx$ **2**
- (e) Evaluate $\int_0^{\frac{\pi}{6}} \cos x \sin^3 x \, dx$ **3**
- (f) For what values of m is the polynomial $x^2 - (m+2)x + 2(m+2)$ positive for all x **2**
- (g) Consider the expansion $(2x-p)^9$. The coefficient of x^6 is -344064 . **2**
Find the value of p

Question 12 (15 marks) Use a SEPARATE writing booklet

- (a) Find $\int \frac{dx}{25+4x^2}$ **2**
- (b) Given that $\sin \theta = \frac{3}{4}$ and $\frac{\pi}{2} \leq \theta \leq \pi$ determine the exact value of $\tan 2\theta$ **3**
- (c) (i) Find the derivative of $x \log_e x - x$ **1**
(ii) Hence evaluate $\int_{\sqrt{e}}^e \log_e x \, dx$ **2**
- (d) A mixed volleyball team of eight players is selected from ten males and nine females **2**
In how many ways can this be done if the team must have at least 2 female players
- (e) The vectors $\vec{u} = \begin{pmatrix} a \\ 3 \end{pmatrix}$ and $\vec{v} = \begin{pmatrix} 2 \\ 7 \end{pmatrix}$ are parallel. Find the value of a **2**
- (f) The polynomial $P(x) = (x-p)^3 + q$ is zero at $x = 1$ and when divided by x **3**
the remainder is -7 . Find the possible values of p

Question 13 (15 marks) Use a SEPARATE writing booklet

(a) Evaluate $\int_{-1}^3 x\sqrt{3-x} \, dx$ using the substitution $u = 3 - x$. 4

(b) (i) Express $\sqrt{3} \cos x + \sin x$ in the form $R \cos(x - \alpha)$ where $R > 0$ and α is acute. 2

(ii) Hence solve $\sqrt{3} \cos x + \sin x = 1$ for $0 \leq x \leq 2\pi$. 2

(c) Use the process of Mathematical induction to prove 3

$2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (n^2 + 1)n! = n(n+1)!$ for all positive integers n .

(d) A population of mice in a meadow after t years satisfies the logistic 4

differential equation $\frac{dP}{dt} = \frac{3P}{2500}(2500 - P)$, where the initial population of mice is 500.

Given $\frac{1}{P} + \frac{1}{2500 - P} = \frac{2500}{P(2500 - P)}$, solve the differential equation to find the

population P of the mice at time t . Express your answer in the form

$P = \frac{A}{1 - Be^{-kt}}$ where A , B and k are integers.

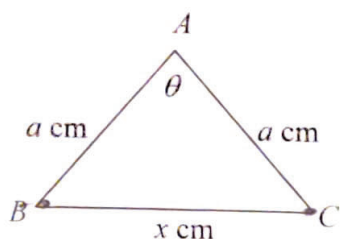
End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) Solve the differential equation $(x^2 + 1) \frac{dy}{dx} = 6xy$ where $x = 1$ and $y = 2$ giving your answer as y in terms of x . 3

- (b) Given that $y = e^{2x} + e^{-2x}$, determine the values of constants a and b that satisfy the following differential equation $\frac{d^2y}{dx^2} + a \frac{dy}{dx} + by = 5e^{2x} + e^{-2x}$. 4

- (c) In the triangle ABC , $AB = AC = a$ cm. The angle BAC is increasing at the rate of 2 radians/min. Let $\angle BAC = \theta$ radians and $BC = x$ cm.



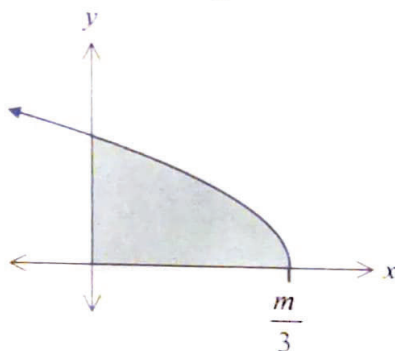
NOT TO
SCALE

- (i) Show that $x = a\sqrt{2 - 2\cos\theta}$. 1

- (ii) Show that $\frac{dx}{d\theta} = \frac{a \sin \theta}{\sqrt{2 - 2\cos \theta}}$. 1

- (iii) Determine, in terms of a , the rate with respect to time at which BC is increasing when $\theta = \frac{\pi}{3}$ radians. 2

- (d) Let $f(x) = \sqrt{m - 3x}$ for $x < \frac{m}{3}$. The graph of $y = f(x)$ is shown. 4



The area enclosed by the graph $y = f(x)$, the x -axis and the y -axis is rotated about the y axis. Find the value of m such that the volume of the solid formed is $\frac{5000\pi}{27}$ cubic units.

End of Paper

-1-

MATHEMATICS EXTENSION 1 TRIAL 2022

1. $\underline{a} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$ $\underline{b} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$

$$\underline{a} \cdot \underline{b} = |\underline{a}| |\underline{b}| \cos \theta$$

$$\cos \theta = \frac{\underline{a} \cdot \underline{b}}{|\underline{a}| |\underline{b}|}$$

$$= \frac{(3 \times -1 + 4 \times 2)}{\sqrt{9+16} \sqrt{1+4}}$$

$$= \frac{5}{5\sqrt{5}} = \frac{1}{\sqrt{5}}$$

$$\theta = \cos^{-1} \frac{1}{\sqrt{5}}$$

$$\approx 63^\circ \quad \textcircled{B}$$

2. $\textcircled{D} \quad \frac{\sin \theta}{\tan \theta} = \frac{\sin \theta}{\frac{\sin \theta}{\cos \theta}}$

$$= \sin \theta \times \frac{\cos \theta}{\sin \theta}$$

$$= \cos \theta$$

3. $P(x) = x^3 - x^2 - 5x - 3$

$$P'(x) = 3x^2 - 2x - 5$$

Double root $\therefore P'(a) = 0$

$$0 = 3a^2 - 2a - 5 \quad \begin{cases} P = -15 \\ S = -2 \\ P = -5.3 \end{cases}$$

$$0 = 3a^2 + 3a - 5a - 5$$

$$0 = 3a(a+1) - 5(a+1)$$

$$0 = (a+1)(3a-5)$$

$$\therefore a = -1 \text{ or } \frac{5}{3}$$

$$P(-1) = (-1)^3 - (-1)^2 - 5(-1) - 3$$

$$= -1 - 1 + 5 - 3$$

$$= 0$$

$$\therefore P(-1) = 0 \quad P'(-1) = 0$$

$$\therefore \textcircled{B}$$

4. $\int \sin x \cos x \, dx = \frac{1}{2} \int \sin 2x \, dx$

$$= \frac{1}{2} \left(-\frac{\cos 2x}{2} \right) + c$$

$$= -\frac{\cos 2x}{4} + c$$

$\textcircled{C}$

5. $-1 \leq 2x + 1 \leq 1$

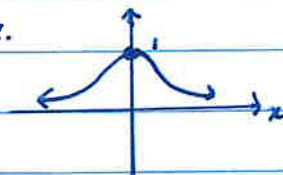
$$-2 \leq 2x \leq 0$$

$$-1 \leq x \leq 0 \quad [-1, 0] \quad \textcircled{A}$$

6. $4 \times 7 + 1 = 29$

$\textcircled{B}$

7.



$$0 < y \leq 1$$

$$(0, 1] \quad \textcircled{C}$$

8. A. $y = \sec \theta = \frac{1}{\cos \theta} = 1$

B. $y = \sec\left(\frac{2\pi}{3}\right)$

C. $y = \cot(0) = \frac{\cos 0}{\sin 0}$ $\sin 0 = 0 \therefore \text{Undefined at } x=0 \quad \textcircled{C}$

9. $\int \frac{4}{\sqrt{3^2 - x^2}} \, dx = 4 \sin^{-1} \frac{x}{3} + c \quad \textcircled{D}$

10. $\textcircled{A} \quad x=1 \quad y=1 \quad \frac{dy}{dx} = 1-1=0 \quad \checkmark$ (A or D)

$$x=-2 \quad y=-1 \quad \frac{dy}{dx} = 1$$

- | | |
|----|---|
| 1 | B |
| 2 | D |
| 3 | B |
| 4 | C |
| 5 | A |
| 6 | B |
| 7 | C |
| 8 | C |
| 9 | D |
| 10 | A |

D. $\frac{dy}{dx} = -2+1 = -1 \quad x$

$\therefore \textcircled{A}$

(15 Marks)

11 a) $\frac{3}{2x-5} \leq -1$

critical values

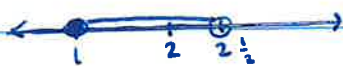
$$x \neq \frac{5}{2}$$

$$\frac{3}{2x-5} = -1$$

$$3 = -2x + 5$$

$$2x = 2$$

$$x = 1$$



$$\text{test } x = 2 \quad \frac{3}{-1} \leq -1 \quad \checkmark$$

$$1 \leq x < 2\frac{1}{2}$$

[3]

b) $y = \cos^{-1} \frac{3x}{2}$

$$\frac{dy}{dx} = \frac{-3}{\sqrt{4-9x^2}}$$

$$\begin{cases} a=2 \\ f(x)=3x \\ f'(x)=3 \end{cases}$$

[1]

c) perpendicular when

$$\begin{pmatrix} m \\ 2m+6 \end{pmatrix} \cdot \begin{pmatrix} m+1 \\ -1 \end{pmatrix} = 0$$

$$m(m+1) + -1(2m+6) = 0$$

$$m^2 + m - 2m - 6 = 0$$

$$m^2 - m - 6 = 0$$

$$(m-3)(m+2) = 0$$

$$\therefore m = 3 \text{ or } -2$$

[2]

d) $\int \cos^2 4x \, dx = \int \left(\frac{1}{2} + \frac{1}{2} \cos 8x \right) dx$

$$= \frac{1}{2}x + \frac{1}{2} \frac{\sin 8x}{8} + c$$

$$= \frac{1}{2}x + \frac{1}{16} \sin 8x + c \quad [2]$$

e) $\int_0^{\frac{\pi}{6}} \cos x \sin^3 x \, dx$

$$= \left[\frac{\sin^4 x}{4} \right]_0^{\frac{\pi}{6}}$$

$$= \frac{(\sin \frac{\pi}{6})^4}{4} - \frac{\sin^4 0}{4}$$

$$= \left(\frac{1}{4} \times \frac{1}{2^4} \right) - 0$$

$$= \frac{1}{64}$$

[3]

f) for positive $\Delta < 0$

$$\Delta = b^2 - 4ac$$

$$= [-(m+2)]^2 - 4(2(m+2))$$

$$= m^2 + 4m + 4 - 8m - 16$$

$$= m^2 - 4m - 12$$

$$= (m-6)(m+2) \quad m = -2, 6$$



$$(m-6)(m+2) < 0 \quad \text{test } m = 0$$

$$-6 \times 2 < 0 \quad \checkmark$$

$$-2 < m < 6$$

[2]

g) coefficient of $x^6 = {}^9C_6 (2x)^6 (-p)^3$

$$-344064 = -5376p^3$$

$$64 = p^3$$

$$p = 4$$

[2]

(15 Marks)

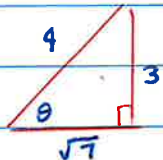
$$12 \text{ a) } \int \frac{dx}{25+4x^2} = \frac{1}{2} \int \frac{2dx}{5^2+(2x)^2}$$

$$\left. \begin{array}{l} a=5 \\ f(x)=2x \\ f'(x)=2 \end{array} \right\}$$

$$= \frac{1}{5} \times \frac{1}{2} \tan^{-1} \frac{2x}{5} + c$$

$$= \frac{1}{10} \tan^{-1} \frac{2x}{5} + c \quad [2]$$

$$b) \sin \theta = \frac{3}{4} \quad (Q2)$$



$$\tan \theta = \frac{3}{4}$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

$$= \frac{2 \times \frac{3}{4}}{1 - (\frac{3}{4})^2}$$

$$= \frac{-6}{\frac{7}{4}}$$

$$= -\frac{6 \times 4}{7} = -\frac{24}{7}$$

$$= \frac{24}{7} \times \frac{\sqrt{7}}{\sqrt{7}}$$

$$= \frac{24\sqrt{7}}{7}$$

$$= 3\sqrt{7}$$

[3]

c)

$$(i) \quad y = x \log_e x - x$$

$$\frac{dy}{dx} = x \times \frac{1}{x} + \log_e x \times 1 - 1$$

$$= 1 + \log_e x - 1$$

$$= \log_e x$$

[1]

$$\left\{ \begin{array}{l} u = x \\ u' = 1 \\ v = \log_e x \\ v' = \frac{1}{x} \end{array} \right.$$

$$(ii) \quad \int_{\sqrt{e}}^e \log_e x \, dx = [x \log_e x - x]_{\sqrt{e}}^e$$

$$= (e \log_e e - e) - (\sqrt{e} \log_e \sqrt{e} - \sqrt{e})$$

$$= (e - e) - (\sqrt{e} \log_e e^{\frac{1}{2}} - \sqrt{e})$$

$$= 0 - (\frac{1}{2} \sqrt{e} \log_e e - \sqrt{e})$$

$$= 0 - \frac{1}{2} \sqrt{e} + \sqrt{e}$$

$$= \frac{1}{2} \sqrt{e}$$

[2]

d) 8 players 10 males 9 females

At least 2 female = Total - (0F + 1F)

$$= {}^{19}C_8 - [{}^9C_0 \times {}^{10}C_8 + {}^9C_1 \times {}^{10}C_7]$$

$$= 74457$$

[2]

2)(e) $u = \begin{pmatrix} a \\ 3 \end{pmatrix}$ $v = \begin{pmatrix} 2 \\ 7 \end{pmatrix}$
For parallel $\begin{pmatrix} a \\ 3 \end{pmatrix} = k \begin{pmatrix} 2 \\ 7 \end{pmatrix}$

$$3 = k \times 7$$

$$\frac{3}{7} = k$$

$$\therefore a = \frac{3}{7} \times 2$$

$$a = \frac{6}{7}$$

[2]

(f) $P(x) = (x-p)^3 + q$
 $P(1) = 0$
 $P(0) = -7$

$$P(1) = (1-p)^3 + q$$

$$0 = (1-p)^3 + q \quad (1)$$

$$P(0) = (-p)^3 + q$$

$$-7 = (-p)^3 + q$$

$$-7 = -p^3 + q$$

$$-7 + p^3 = q \quad (2)$$

Sub (2) into (1)

$$0 = (1-p)^3 - 7 + p^3$$

$$0 = 1 - 3p + 3p^2 - p^3 - 7 + p^3$$

$$0 = 3p^2 - 3p - 6$$

$$0 = p^2 - p - 2$$

$$0 = (p-2)(p+1)$$

$$\therefore p = 2 \text{ or } -1$$

[3]

(13) $\int_{-1}^3 x \sqrt{3-x} dx$

$$= - \int_4^0 (3-u) u^{\frac{1}{2}} du$$

$$= \int_0^4 (3u^{\frac{1}{2}} - u^{\frac{3}{2}}) du$$

$$= \left[\frac{3u^{\frac{3}{2}}}{\frac{3}{2}} - \frac{u^{\frac{5}{2}}}{\frac{5}{2}} \right]_0^4$$

$$= \left[2u^{\frac{3}{2}} - \frac{2}{5}u^{\frac{5}{2}} \right]_0^4$$

$$= \left[2(4)^{\frac{3}{2}} - \frac{2}{5}(4)^{\frac{5}{2}} \right] - [0-0]$$

$$= 16 - \frac{64}{5}$$

$$= \frac{16}{5}$$

[4]

b)(i) $\sqrt{3} \cos x + \sin x = R \cos x \cos \alpha + R \sin x \sin \alpha$

$$R \cos \alpha = \sqrt{3}$$

$$R \sin \alpha = 1$$

$$R = \sqrt{3+1} = 2$$

$$\cos \alpha = \frac{\sqrt{3}}{2}$$

$$\sin \alpha = \frac{1}{2}$$

$$\tan \alpha = \frac{1}{\sqrt{3}}$$

$$\alpha = \frac{\pi}{6}$$

$$\therefore \sqrt{3} \cos x + \sin x = 2 \cos(x - \frac{\pi}{6})$$

[2]

(ii) $2 \cos(x - \frac{\pi}{6}) = 1$

$$\cos(x - \frac{\pi}{6}) = \frac{1}{2}$$

$$\checkmark \text{ ref } \alpha = \frac{\pi}{3}$$

$$\therefore x - \frac{\pi}{6} = \frac{\pi}{3} \text{ or } x - \frac{\pi}{6} = \frac{5\pi}{3}$$

$$x = \frac{\pi}{3} + \frac{\pi}{6}$$

$$x = \frac{5\pi}{3} + \frac{\pi}{6}$$

$$= \frac{\pi}{2} \text{ or }$$

$$= \frac{11\pi}{6}$$

[2]

(c) step 1: Prove true for $n=1$

$$\begin{aligned} \text{LHS} &= 2 \times 1! \\ &= 2 \end{aligned}$$

$$\begin{aligned} \text{RHS} &= 1(1+1)! \\ &= 2! \\ &= 2 \end{aligned}$$

$$\therefore \text{LHS} = \text{RHS}$$

True for $n=1$

step 2: Assume true for $n=k$

$$2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (k^2 + 1)k! = k(k+1)! \quad *$$

step 3: Prove true for $n=k+1$

$$\text{RTP: } \underbrace{2 \times 1! + 5 \times 2! + 10 \times 3! + \dots + (k^2 + 1)k!}_{S_k} + [(k+1)^2 + 1](k+1)! = (k+1)(k+2)!$$

$$\begin{aligned} \text{LHS} &= k(k+1)! + [k^2 + 2k + 1 + 1](k+1)! \quad (\text{by assumption}) \\ &= (k+1)! [k + k^2 + 2k + 2] \\ &= (k+1)! [k^2 + 3k + 2] \\ &= (k+1)! (k+2)(k+1) \\ &= \underbrace{(k+1)! (k+2)}_{(k+2)!} (k+1) \\ &= \text{RHS} \end{aligned}$$

step 4: Since true for $n=1$, then true for $n=2, n=3$ and so on for all positive integers.

[3]

$$\text{cd) } \frac{dp}{dt} = \frac{3p}{2500} (2500 - p)$$

Trivial solns: $p = 0$
 $p = 2500$

$$\frac{dp}{\frac{p(2500-p)}{2500}} = 3 dt$$

$$\int \frac{2500}{p(2500-p)} dp = \int 3 dt$$

$$\int \left(\frac{1}{p} + \frac{1}{2500-p} \right) dp = 3 \int dt$$

$$\ln |p| - \ln |2500-p| = 3t + C$$

$$\ln \left| \frac{p}{2500-p} \right| = 3t + C$$

$$\left| \frac{p}{2500-p} \right| = e^{3t+C}$$

$$= e^{3t} \times e^C$$

$$= A e^{3t}$$

where $A = e^C$

$$\frac{p}{2500-p} = A e^{3t}$$

$$A = \pm e^C$$

when $t = 0$
 $p = 500$

$$\frac{500}{2000} = A e^0$$

$$\frac{1}{4} = A$$

$$\therefore \frac{p}{2500-p} = \frac{1}{4} e^{3t}$$

$$4p = e^{3t} (2500 - p)$$

$$4p = 2500 e^{3t} - p e^{3t}$$

$$4p + p e^{3t} = 2500 e^{3t}$$

$$p(4 + e^{3t}) = 2500 e^{3t}$$

$$p = \frac{2500 e^{3t}}{4 + e^{3t}} \div e^{3t}$$

$$= \frac{2500}{\frac{4}{e^{3t}} + 1}$$

$$= \frac{2500}{1 + 4e^{-3t}}$$

$$B = -4$$

[4]

or $p = \frac{2500}{1 - (-4)e^{-3t}}$

(15 Marks)

-7-

14 $(x^2+1) \frac{dy}{dx} = 6xy$

a) $\int \frac{dy}{y} = \int \frac{6x}{x^2+1} dx$

$$\ln |y| = 3 \int \frac{2x}{x^2+1} dx$$

$$\ln |y| = 3 \ln |x^2+1| + c$$

$$x=1 \quad y=2 \quad \ln 2 = 3 \ln 2 + c$$

$$-2 \ln 2 = c$$

$$\therefore \ln y = 3 \ln (x^2+1) - 2 \ln 2$$

$$\ln y = \ln (x^2+1)^3 - \ln 2^2$$

$$y = \frac{(x^2+1)^3}{4}$$

[3]

b) $y = e^{2x} + e^{-2x}$

$$\frac{dy}{dx} = 2e^{2x} - 2e^{-2x}$$

$$\frac{d^2y}{dx^2} = 4e^{2x} + 4e^{-2x}$$

$$\begin{aligned} \frac{d^2y}{dx^2} + a \frac{dy}{dx} + by &= 4e^{2x} + 4e^{-2x} + 2ae^{2x} - 2ae^{-2x} + be^{2x} + be^{-2x} \\ &= (4+2a+b)e^{2x} + (4-2a+b)e^{-2x} \end{aligned}$$

Equating coefficients:

$$4 + 2a + b = 5$$

$$4 - 2a + b = 1$$

$$2a + b = 1 \quad (1)$$

$$-2a + b = -3 \quad (2)$$

$$+ \quad -2a + b = -3 \quad (2)$$

$$-2a - 1 = -3$$

$$2b = -2$$

$$-2a = -2$$

$$b = -1$$

$$a = 1$$

$$\therefore \underline{a=1 \quad b=-1}$$

[4]

(i) cosine rule:

$$c^2 = a^2 + b^2 - 2ab \cos c$$

(ii) $x^2 = a^2 + a^2 - 2a^2 \cos \theta$

$$= 2a^2 - 2a^2 \cos \theta \quad (\text{factorise } a^2)$$

$$\underline{x = a \sqrt{2 - 2 \cos \theta}} \quad \text{which is required} \quad [1]$$

ii) $\frac{dx}{d\theta} = a \times \frac{1}{2} (2 - 2 \cos \theta)^{-\frac{1}{2}} \times 2 \sin \theta$

$$= \frac{a \sin \theta}{\sqrt{2 - 2 \cos \theta}}$$

which is required

[1]

(iii) $\frac{dx}{dt} = \frac{dx}{d\theta} \times \frac{d\theta}{dt}$

$$= \frac{a \sin \theta}{\sqrt{2 - 2 \cos \theta}} \times 2 \quad \text{when } \theta = \frac{\pi}{3}$$

$$= \frac{2a \sin \frac{\pi}{3}}{\sqrt{2 - 2 \cos \frac{\pi}{3}}}$$

$$= \frac{2a \frac{\sqrt{3}}{2}}{\sqrt{1}}$$

$$= \underline{\sqrt{3}a} \text{ cm/min}$$

[2]

d) $V = \pi \int_0^m x^2 dy$

$$= \pi \int_0^m (y^4 - 2my^2 + m^2) dy$$

$$= \frac{\pi}{9} \left[\frac{y^5}{5} - 2m \frac{y^3}{3} + m^2 y \right]_0^m$$

$$= \frac{\pi}{9} \left[\frac{(\sqrt{m})^5}{5} - \frac{2m}{3} (\sqrt{m})^3 + m^2 \sqrt{m} \right]$$

$$= \frac{\pi}{9} \left[\frac{m^2 \sqrt{m}}{5} - \frac{2}{3} m^2 \sqrt{m} + m^2 \sqrt{m} \right]$$

$$= \frac{\pi}{9} \times m^2 \sqrt{m} \left(\frac{1}{5} - \frac{2}{3} + 1 \right)$$

$$= m^2 \sqrt{m} \frac{\pi}{9} \times \frac{8}{15}$$

$$= \underline{\underline{\frac{8}{135} m^2 \sqrt{m} \pi}}$$

$$y = \sqrt{m - 3x} \quad x=0 \quad y=\sqrt{m}$$

$$y^2 = m - 3x$$

$$3x = m - y^2$$

$$x = \frac{m - y^2}{3}$$

$$x^2 = \frac{(m - y^2)^2}{9}$$

$$= \frac{m^2 - 2my^2 + y^4}{9}$$

$$\frac{8}{135} m^{5/2} \pi = \frac{5000 \pi}{27}$$

$$m^{5/2} = 3125$$

$$\underline{m = 25}$$

[4]